

Project Scout

Master Project Summary

Last updated: June 8, 2026 | Version 20

Upload this document at the start of every new Claude or ChatGPT conversation about Scout. This is the single source of truth for Scout's identity, architecture, business model, and current state.

June 8, 2026 — What Changed Since June 7:

- ScoutFaceView thinking-state expression session completed.
- Iris freeze during thinking FIXED — eyes now drift gently instead of locking dead-center.
- Iris jumping calmed — saccade slowed during thinking (τ 80f $\rightarrow$ 600f) so it can't snap.
- Thinking gaze: eyes drift gently up and around center (thinkGazeY), not frozen.
- Lower eyelid raised while thinking (26f $\rightarrow$ 34f) — narrowed, focused look.
- Upper eyelid lowered slightly while thinking (+0.22) — concentration.
- Asymmetric thinking brows — left brow lifts strongly (~34f), right brow stays low. Reads as 'hmm, thinking.'
- Inner brow knit (downward tilt toward nose) attempted but did not read on-device — PARKED for a future pass.

1. Who We Are

Patrick Lippy — developer, project owner, creator of Scout. Not a professional programmer. Stroke survivor, blind in right eye, type 1 diabetic, dyslexic. Explain things calmly and at screenshot level.

- Diana: Patrick's wife
- Elijah: Patrick's son, age 9. Scout's biggest fan. Has drawn pictures of Scout.
- Nicolas: The family dog. The Nicolas Protocol: Scout stops immediately when the dog is detected.

Names must NEVER be hardcoded in Scout's code. Always use variables.

AI Collaborators: Patrick works with both Claude and ChatGPT as project partners. Cross-review between the two is welcome and encouraged. Grok was tried and discontinued.

2. What Scout Is

Scout is a calm family companion robot running on a Samsung Galaxy phone mounted in landscape mode as a permanent face display. Scout has animated eyes, speaks, listens, sees via camera, and remembers the family.

- Package: com.example.scoutface | Language: Kotlin + C++ NDK
- Primary device: Samsung Galaxy A32 — stress-test device, not the benchmark
- Dev device: Samsung Galaxy Fold 7 (wireless via WiFi from Android Studio)
- Future hardware: KEYESTUDIO Mini Tank Kit V2 chassis via Bluetooth (opt-in)
- Ship target: Google Play Store — \$9.99 one-time purchase. No subscriptions. Ever.

3. Identity & Purpose

Scout should feel: Calm. Thoughtful. Emotionally subtle. Grounded. Occasionally curious. Sometimes unsure. Quietly alive. Honest. Predictable. Present.

Scout should NOT: Constantly praise the user. Act overly excited. Feel fake or scripted. Behave unpredictably. Constantly force conversation. Use permanent goodbye language.

Core Philosophy: Stability > Features | Presence > Intelligence | Honest > Fake cheerful | Predictable > Flashy | Local-first > Cloud dependence

4. Business Model

App: \$9.99 one-time purchase on Google Play Store. No recurring fees. No subscriptions. Ever. This is the cornerstone of Scout's value. Every architectural decision supports this principle.

Baseline Brain: TinyLlama 1.1B Chat Q4_K_M (~669 MB) ships with every Scout via Google Play Asset Delivery. Runs entirely offline. VERIFIED on BOTH Galaxy A32 and Fold 7.

Optional Gemini: Users add their own free Gemini key in Settings. Scout NEVER ships with a bundled key — every user uses their own quota.

5. Support Scout (In-App, Optional)

A 'Support Scout' screen offers optional one-time contributions. Scout is a one-time purchase; support is completely optional and never unlocks core features. Scout will always stay ad-free.

Tier	Amount	Label
Coffee	\$3	Buy Scout a Coffee
More	\$5	Support Scout More
Grow	\$10	Help Scout Grow
Founding	\$25	Founding Supporter

Decision (June 4, 2026): The \$25 Founding Supporter tier stays. It is not for everyone — it is for the person who connects deeply with Scout and wants to be part of the story.

6. Architecture — Five-Layer Memory Stack

Layer	Type	Storage	Status
Working	Sensory	RAM	Done
Habit	Patterns	JSON 14-day	Done
Truth	Authority	SQLite	Done
Relevance	Index	Local Vector	Not yet
Reflective	Wisdom	LLM read-only	Not yet

- Sovereign Rules: SQLite Truth always overrules everything.
- Privacy Gate: Gemini receives anonymized text only (planned).
- Nicolas Protocol: dog detection → immediate stop.
- Guest Mode: unknown face → 'Hello, I am Scout. What is your name?'
- Creator Memory: CREATOR fact slot planned for TruthDb — who built Scout.

7. Current Technical State

Working:

- ✓ Animated face (ScoutFaceView) — mouth oscillation fixed May 31
- ✓ Speech recognition (Android STT) + Text to Speech (TTS)
- ✓ Camera — face detection (ML Kit), scene labeling
- ✓ Gemini API integration (gemini-3.5-flash)
- ✓ Memory layers: TruthDb, ConversationDb, HabitLayer, PeopleDb, JournalDb
- ✓ Intent router — time, date, greetings, family facts, downloads, vision, weather, IDENTITY
- ✓ Teaching system: 'My name is Patrick' → stored permanently
- ✓ Weather — current, tonight, tomorrow, specific day, 7-day, precipitation % via Open-Meteo
- ✓ ScoutPresenceDecider — four time-of-day modes
- ✓ OFFLINE BRAIN COMPLETE — TinyLlama 1.1B on BOTH A32 and Fold 7
- ✓ TinyLlama crash on A32 stabilized (June 7)
- ✓ Identity questions hardcoded in intent router (June 5)
- ✓ Identity routing expanded: chatbot, feelings, creator, current activity (June 7)
- ✓ Total offline mode — 'go offline' blocks ALL internet features (June 5)
- ✓ Thinking-state expression: gentle iris drift, narrowed lids, asymmetric brows (NEW — June 8)

Pending — Do This First:

- ScoutFaceView — inner brow knit (downward tilt toward nose) while thinking; didn't read on-device June 8. Revisit.
- ScoutFaceView — iris jitter on A32 idle (separate from thinking, hardware timing). Fold 7 smooth.
- Creator memory — implement CREATOR fact slot in TruthDb. (NEXT MAJOR TASK)
- STT phonetic/fuzzy matching — 'Scout' misheard as 'Gal', 'Scott', 'Out'.
- Downloads should respect 'go offline' (finish total-offline mode).

- Response cleanup layer — post-TinyLlama filter before TTS (safety + identity + honest news handling).
- Scout news feed (FUTURE FEATURE) — small live news fetcher, same clean pattern as Open-Meteo.
- MainActivity cleanup and shortening — ongoing.
- Offline Brain Phase 3 — Google Play Asset Delivery for TinyLlama and brain packs.
- Wire in full mood system (CALM/CURIOUS/HAPPY/THINKING/CONCERNED) — thinking pose now partly built.
- Verify WordNet/idioms/sentiment usage — downloaded but may not be actively queried.

7b. b8946 API Discoveries — CRITICAL

Hard-won lessons from getting TinyLlama running on the A32. Future sessions must respect these or Scout will crash.

llama_vocab is a separate type: In b8946, vocabulary was split from the model into a separate opaque type. llama_tokenize and llama_token_to_piece now require const llama_vocab*, NOT model*. Get vocab via llama_model_get_vocab(model) at start of generation.

Functions that crash — hardcode instead: llama_n_vocab(model) → hardcode n_vocab = 32000. llama_token_eos(model) → hardcode eos = 2. llama_token_eot(model) → hardcode eot = 2.

KV Cache / Context Recreation: llama_kv_cache_clear(ctx) does not exist. Workaround: call llama_free(sm->ctx) and recreate context with llama_new_context_with_model at start of every nativeGenerate call. Adds ~300ms but guarantees clean KV cache.

Logits indexing: After prefill decode, use logits at n_prompt - 1, not 0. In generation loop single-token batches, use index 0.

Backend loading on Android: ggml_backend_load_all() always returns 0 on Android. Use ggml_backend_load(cpuPath) with explicit .so path from dladdr(llama_backend_init).

Struct padding: llama_model_params and llama_context_params have _pad[508] trailing array. Do NOT remove this padding.

Tuning (A32, May 29): n_threads 2 (memory-bound, extra cores don't help). nPredict 64. History 2 turns. ~15 tok/s prefill, ~4 tok/s generation. A32 will always be 20–40s per answer — acceptable, Gemini is fast path.

7c. Known Issues — Do Not Touch Without Discussion

Iris jitter (A32 idle): Animation timing issue, NOT a thread problem. NOT present on Fold 7 (hardware). Separate from the thinking-state drift, which is now fixed.

Barge-in: Deliberately disabled. Runaway loop when Scout's mic picks up his own voice. Status: PARKED.

STT name recognition: 'Scout' misheard as 'Gal', 'Scott', 'Out'. STT/phonetic issue, not TinyLlama. Separate session.

Live news: Neither Gemini nor TinyLlama reads live news. Offline, Scout returns old frozen training-data news. A future news feed is needed.

WordNet/files: english-wordnet zip still zipped. idioms.json, positive/negative-words.txt may not be queried. Verify in code.

7d. Session Log

June 5 — Identity, Weather, Total Offline Mode: Added IDENTITY intent + handleIdentityIntent() for 'are you my friend / can you hear me / are you happy'. Fixed duplicate few-shot answers. Precipitation % and today's forecast working. Offline weather message standardized; stale cache no longer served. DECISION: 'go offline' = totally offline. Confirmed TinyLlama reads frozen training data offline (old COVID news = proof). Added Scout news feed as a future feature.

June 7 — Stability, Identity Expansion, Face Direction: TinyLlama A32 crash stabilized. Identity routing expanded (chatbot, feelings, creator, current activity). Thinking-state gaze stabilization added. Creator memory design started. Face evolution roadmap locked (see Section 10).

June 8 — Thinking Expression (ScoutFaceView): Fixed iris freeze during thinking (gentle drift). Calmed iris jumping (slow saccade while thinking). Raised lower lid (26f→34f) and lowered upper lid (+0.22) while thinking for a focused look. Added asymmetric thinking brows — one lifts, one stays low. Inner brow knit attempted but did not read on-device; parked. All changes built clean and tested on-device.

8. Working Rules — Always Apply

- Full paste-ready replacements only, one file at a time. No snippets. No partial files.
- Surgical CTRL-F (find) and CTRL-R (find and replace) approved for large files.
- Always specify which file tab to click before searching. Wrong tab = wasted turns.
- Some Scout files have NO indentation (flush-left). When a search fails, try a shorter unique single-line search string.
- Some logic (e.g. lowerLidTarget) exists in TWO places — updateLife AND scheduleNextFrame. Change BOTH so they match, or Scout flickers.
- One safe change at a time. Build and test before the next change.
- Never touch speech, camera, or download systems without explicit discussion.
- Never touch ScoutFaceView casually — it is Scout's visual heart.
- Both Claude and ChatGPT are active collaborators — cross-review welcome.
- Patrick is not a professional programmer — explain at screenshot level always.
- Patrick is dyslexic, stroke survivor, blind in right eye, T1 diabetic — keep messages clear, concise, not overwhelming.

9. Key Files

File	Description
MainActivity.kt	Main app — all logic, inner classes, brain. IntentType enum + handleIdentityIntent.
ScoutFaceView.kt	Custom face canvas — all visual animation. Thinking expression updated June 8.
GeminiClient.kt	Gemini HTTP wrapper with cooldown discipline
ScoutPromptBuilder.kt	Builds Gemini system instruction
ScoutGeminiManager.kt	Gemini orchestration
ScoutWeatherManager.kt	Live weather via Open-Meteo — precip %, offline-aware
ScoutPresenceDecider.kt	Social timing layer
ScoutIntentRouter.kt	Intent type routing — includes IDENTITY questions
LlamaEngine.kt	Offline brain JNI wrapper — WORKING
OfflinePromptBuilder.kt	TinyLlama prompt formatter — needs update per brain model
scout_llama_jni.cpp	C++ JNI bridge — compiled into libscout_llama.so
scout_llama_api.h	Self-contained b8946 declarations
CMakeLists.txt	NDK build config
TeachExtractor.kt	Extracts facts like names from speech
HabitLayer.kt	Pattern memory — 14-day decay

10. Scout Animation Goal & Mood System

Scout's face should feel alive and emotionally present, but always calm. Tiny layered movements suggest thinking and presence — not cartoon performance. Scout should NEVER have perfectly still eyes or a static mouth.

Mood System — Designed, Partly Built:

- CALM — Eyes center, brows neutral, subtle smile (0.15). Scout's default.
- CURIOUS — Eyes shift right and up, one brow lifts (0.18), slight tilt (0.12).
- HAPPY — Eyes center, both brows lift gently (0.12), smile increases (0.35).
- THINKING — Eyes drift up/around, lids narrow, brows asymmetric (one up, one low). PARTLY BUILT June 8.
- CONCERNED — Eyes look slightly down, inner brows lift (0.05), tilt inward (-0.18).

Face Evolution Roadmap (Locked June 7, 2026):

Version	What Changes
v1.0	Current eyes. Stable irises. Better identity. Better memory. Fix iris jitter.
v1.5	Eyebrows move per mood. Smooth natural blink every 4–8 seconds.
v2.0	Simple upper eyelids + simple lower eyelids. Still cartoon. Still Scout.
v3.0	Full emotion system. Scout at most expressive. Still obviously Scout.

- Keep forever: blue iris, white sclera, cartoon style. These are Scout's visual identity.
- Add eventually: simple upper eyelids, simple lower eyelids, eyebrows, smooth blink.
- Never add: tear ducts, skin folds, eyelashes, realistic anatomy.
- Design principle: 'Scout stays Scout. He just gets a little more alive.'

11. TinyLlama Safety Architecture

- Layer 1: Hard-coded safety rules — app level, always enforced
- Layer 2: Scout identity rules — system prompt + few-shot examples + hardcoded intents
- Layer 3: TinyLlama response generation — last line of defense only

Critical: TinyLlama was trained on internet text and knows profanity and scary content. Its judgment is inconsistent. Scout must NOT rely on TinyLlama to make correct safety decisions. Future: response cleanup layer runs after TinyLlama, before TTS — removes profanity, blocks unsafe phrases, shortens long replies.

12. Device Requirements & Play Store Expectations

Minimum: Android 13+, 4 GB RAM (6 GB recommended), ~3–5 GB free. Face/voice/memory/offline AI all work; responses slower.

Recommended: Android 13+, 8 GB+ RAM, modern mid-range or flagship CPU (2023+), 10 GB+ free. Faster responses, smoother animations.

Play Store wording (agreed): 'Offline AI performance varies by device. Scout runs on many Android devices, but newer phones and tablets provide faster response times and smoother animations.'

13. Brain Upgrade Models

All models already downloaded to: Internal Storage > Scout > Downloaded_Offline_Batch > Future_Upgrades. GGUF format, work with existing NDK bridge. Each needs a prompt-format update in OfflinePromptBuilder.kt when integrated.

Tier	Size	Model
FREE	669 MB	TinyLlama 1.1B Chat Q4_K_M — baseline, both devices, always included
\$2.99 Pack 1	1.79 GB	Phi-2 2.7B Q4_K_M — Microsoft. Solid first upgrade.
\$4.99 Pack 2	2.02 GB	Llama 3.2 3B Instruct Q4_K_M — Meta. THE most important upgrade.
\$6.99 Pack 3	~2.4 GB	Phi-4 Mini Instruct Q4_K_M — Microsoft (PRIMARY).
\$6.99 Pack 3 ALT	2.39 GB	Phi-3.1 Mini 4K Instruct Q4_K_M — tested backup.
\$9.99 Pack 4	4.92 GB	Llama 3.1 8B Instruct Q4_K_M — Meta. Flagship, Fold 7 class (8 GB+ RAM).

Files to delete: onnxruntime-linux-aarch64-1.26.0.tgz (wrong platform). tinyllama...Q4_K_M (1).gguf (old duplicate that caused the Fold 7 problem). Q5_K_M is a marginal same-model improvement — keep or delete, your choice.

14. Hardware Direction — Optional

Scout works fully without hardware. Reference: KEYESTUDIO Mini Tank Kit V2 (Patrick owns one). Tank chassis, treads, ultrasonic sensors, 8x16 LED panel, Bluetooth. Hardware Mode is opt-in. The \$9.99 baseline must work fully on the phone alone.

15. Future Vision — Scout Writes His Own Behavioral Code

A long-term goal. Scout notices patterns, generates a suggestion for a new behavior, and asks permission before activating it. Nothing changes without explicit consent. Scout cannot write compiled Kotlin — but CAN generate behavioral scripts, response patterns, routing rules, and habit triggers within the existing safe framework.